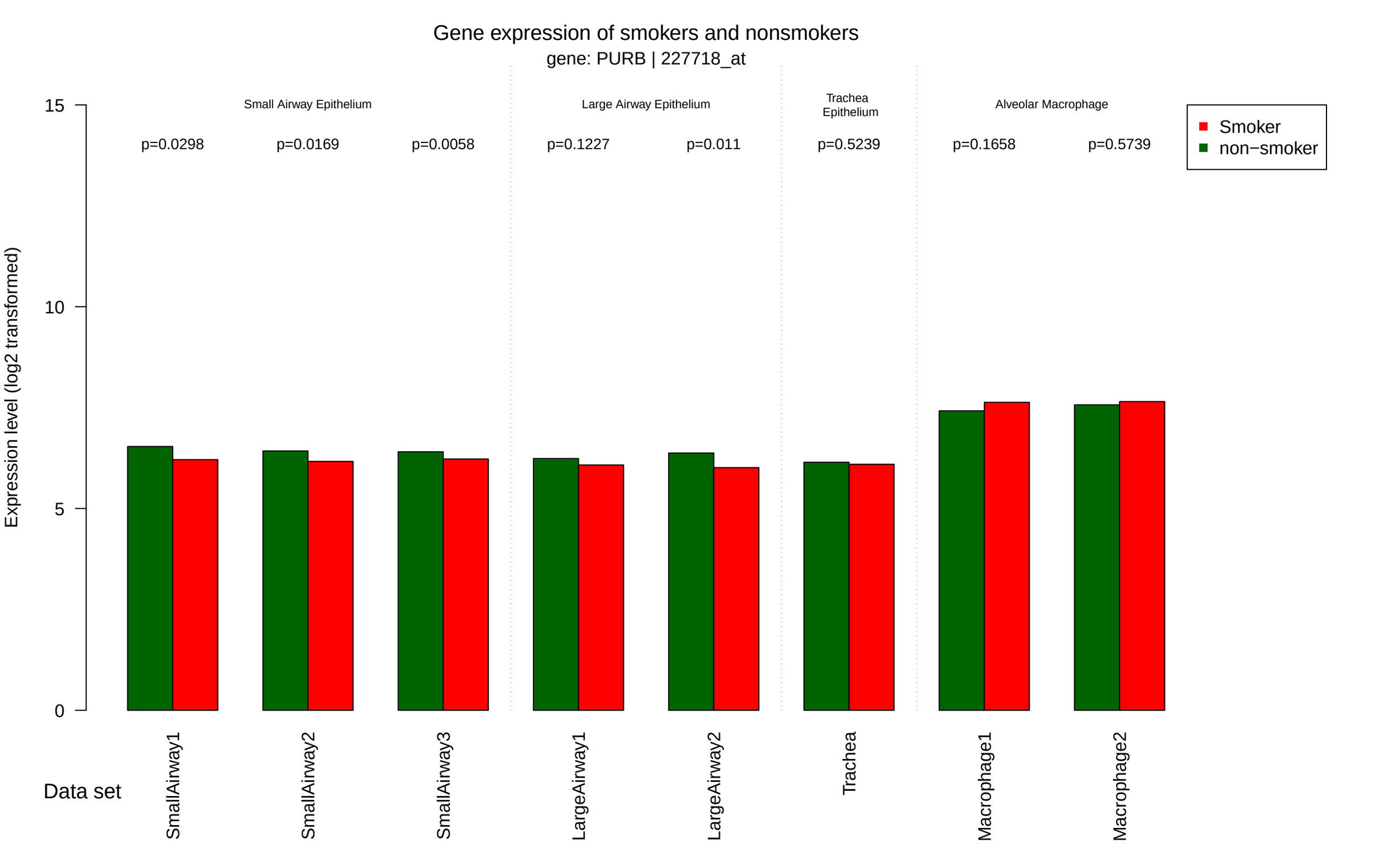
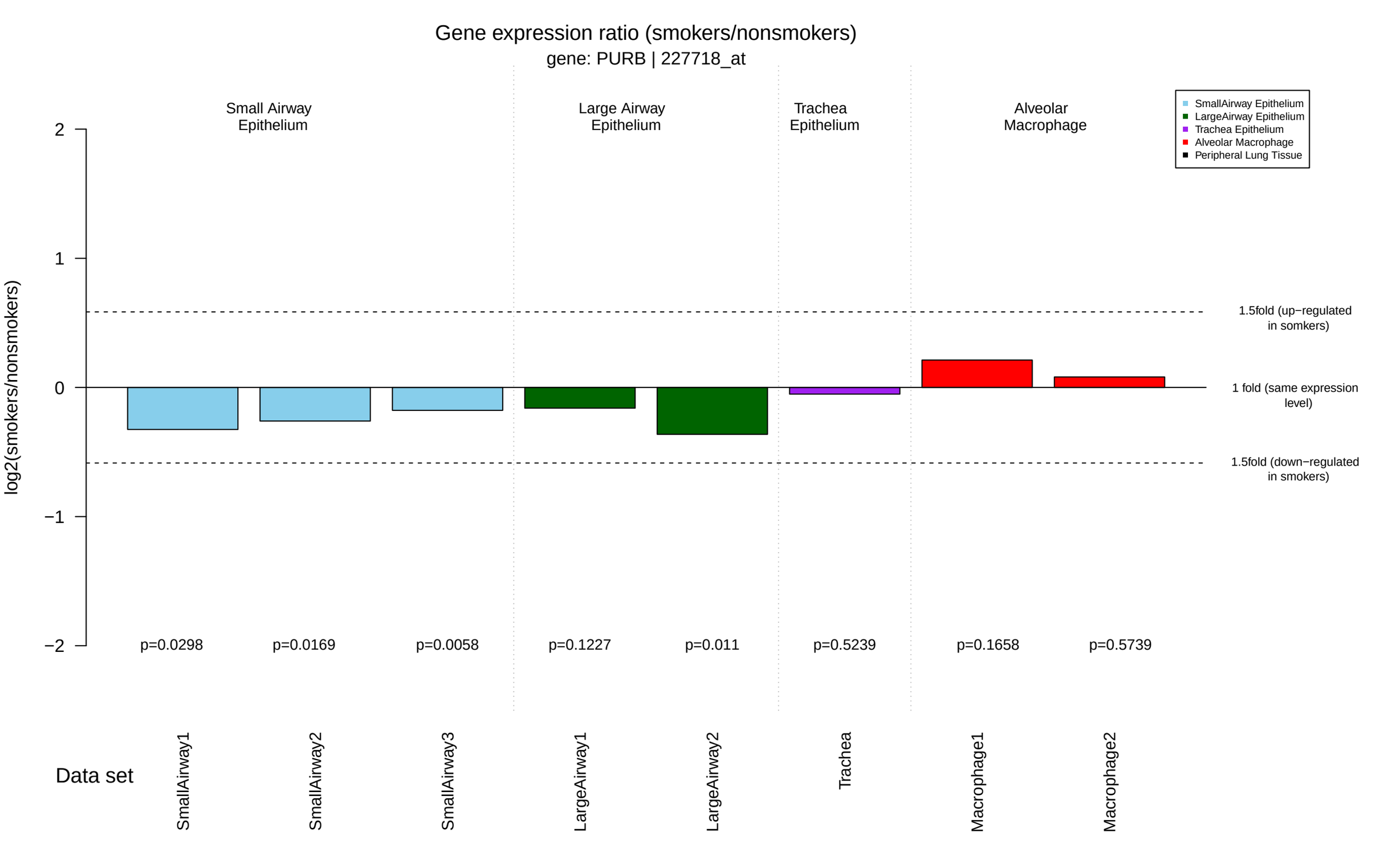
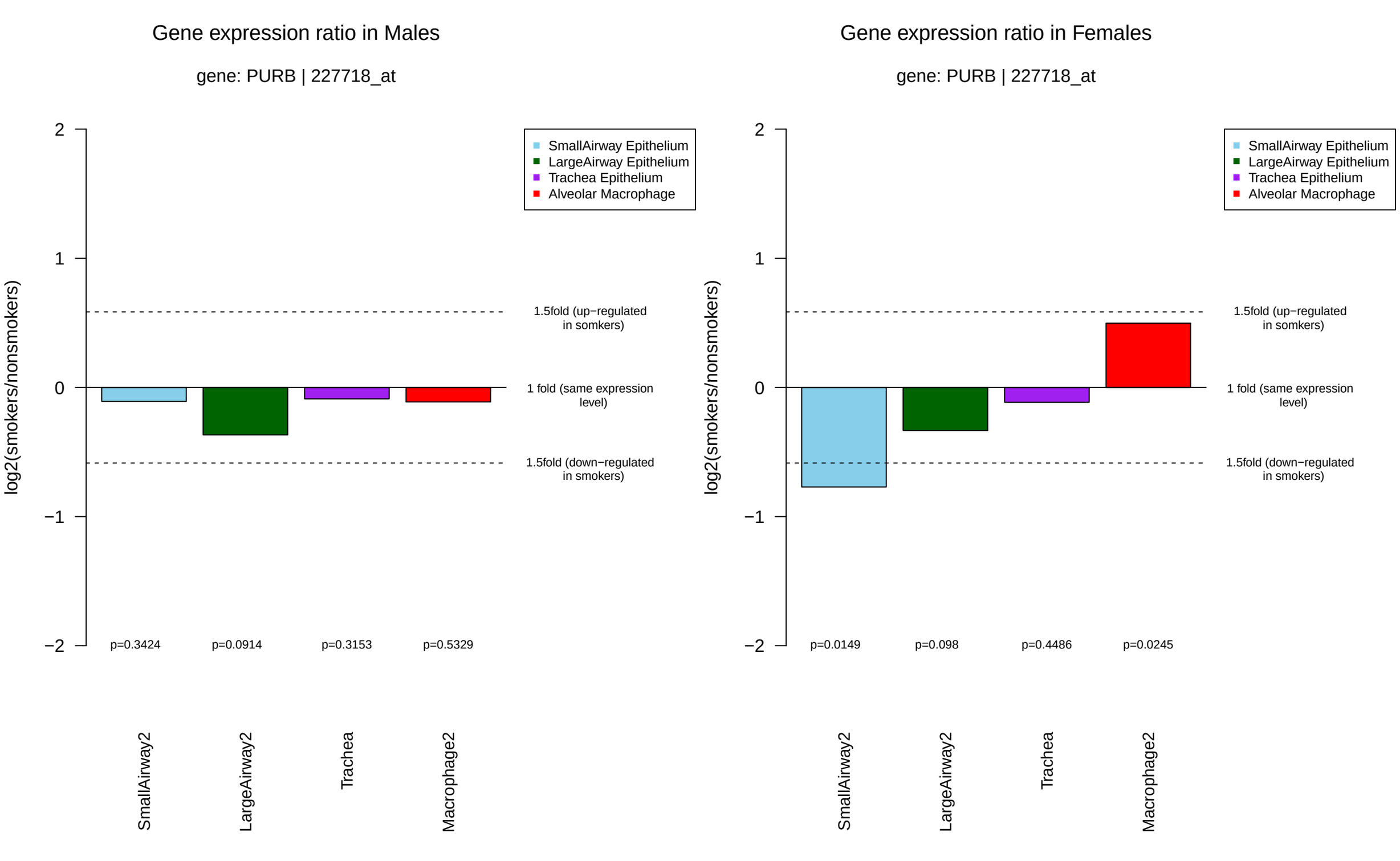
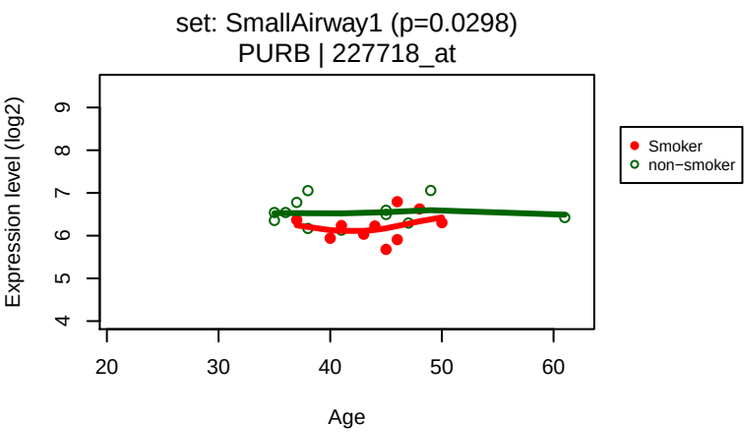
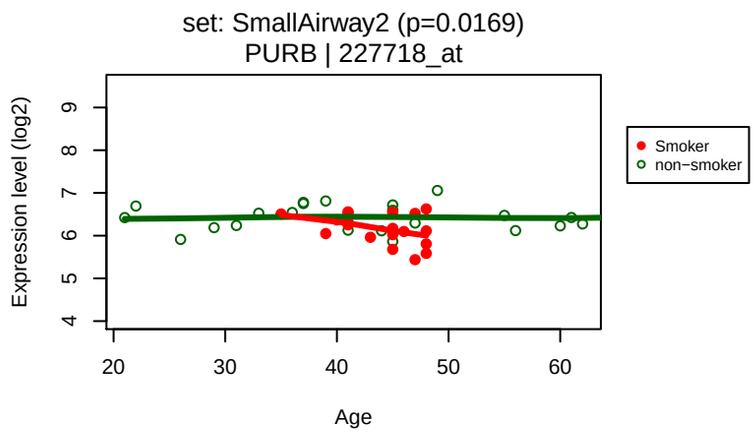
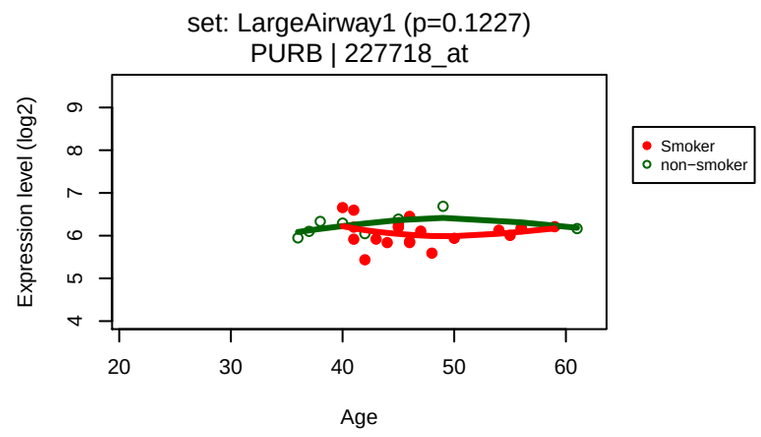

Gene expression of smokers and nonsmokers
gene: PURB | 227718_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0298
p=0.0169
p=0.0058
p=0.1227
p=0.011
p=0.5239
p=0.1658
p=0.5739
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

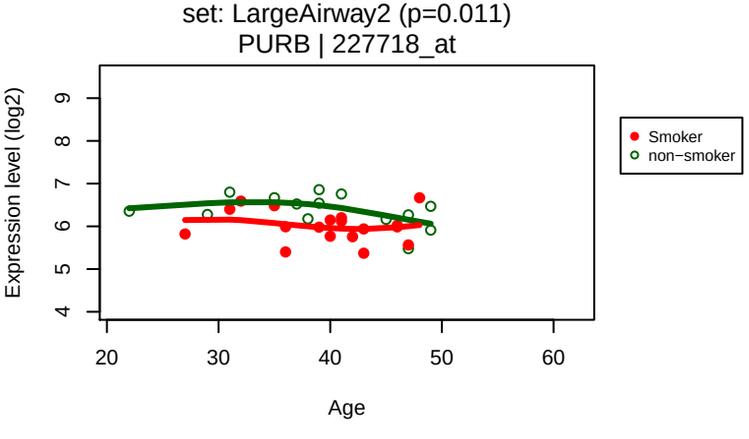

Gene expression ratio (smokers/nonsmokers)
gene: PURB | 227718_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0298
p=0.0169
p=0.0058
p=0.1227
p=0.011
p=0.5239
p=0.1658
p=0.5739
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

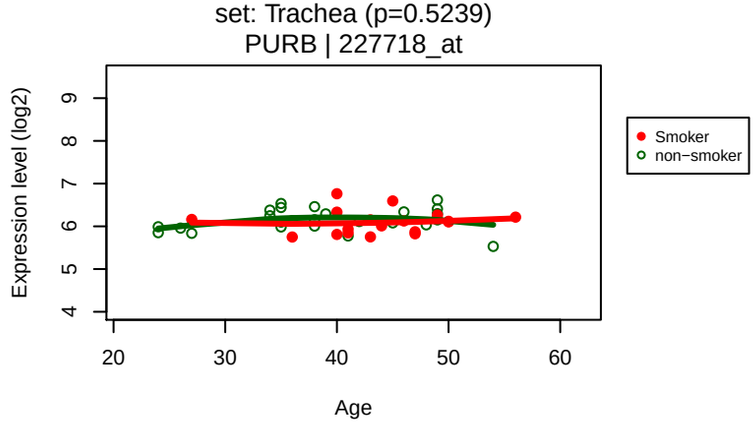

Gene expression ratio in Males
gene: PURB | 227718_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3424
p=0.0914
p=0.3153
p=0.5329
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PURB | 227718_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0149
p=0.098
p=0.4486
p=0.0245
SmallAirway2
LargeAirway2
Trachea
Macrophage2

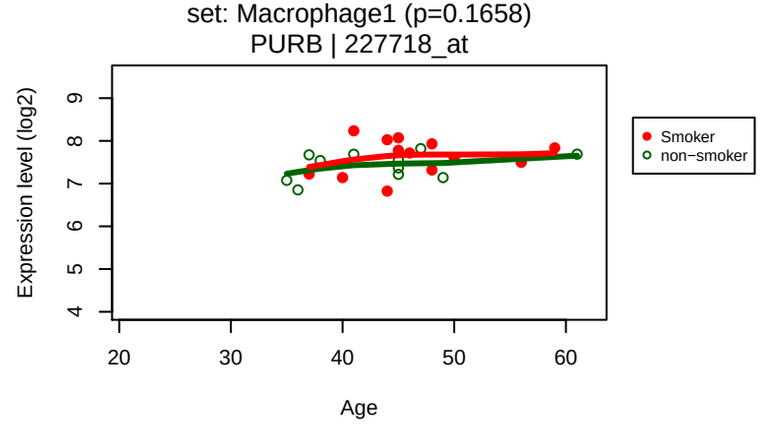

set: SmallAirway1 (p=0.0298)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker

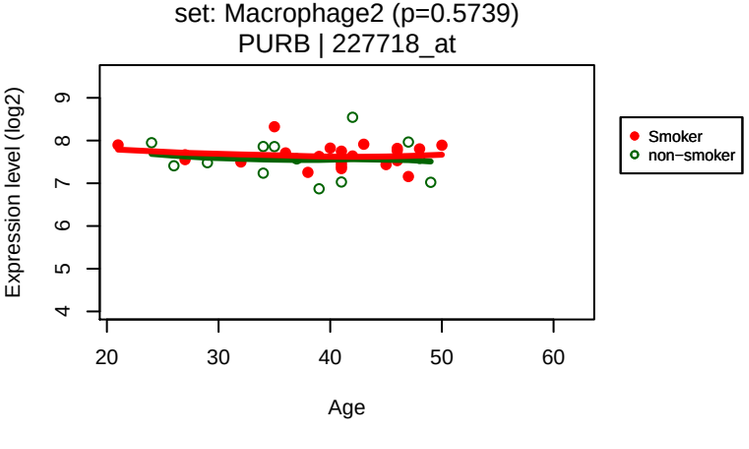

set: SmallAirway2 (p=0.0169)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker

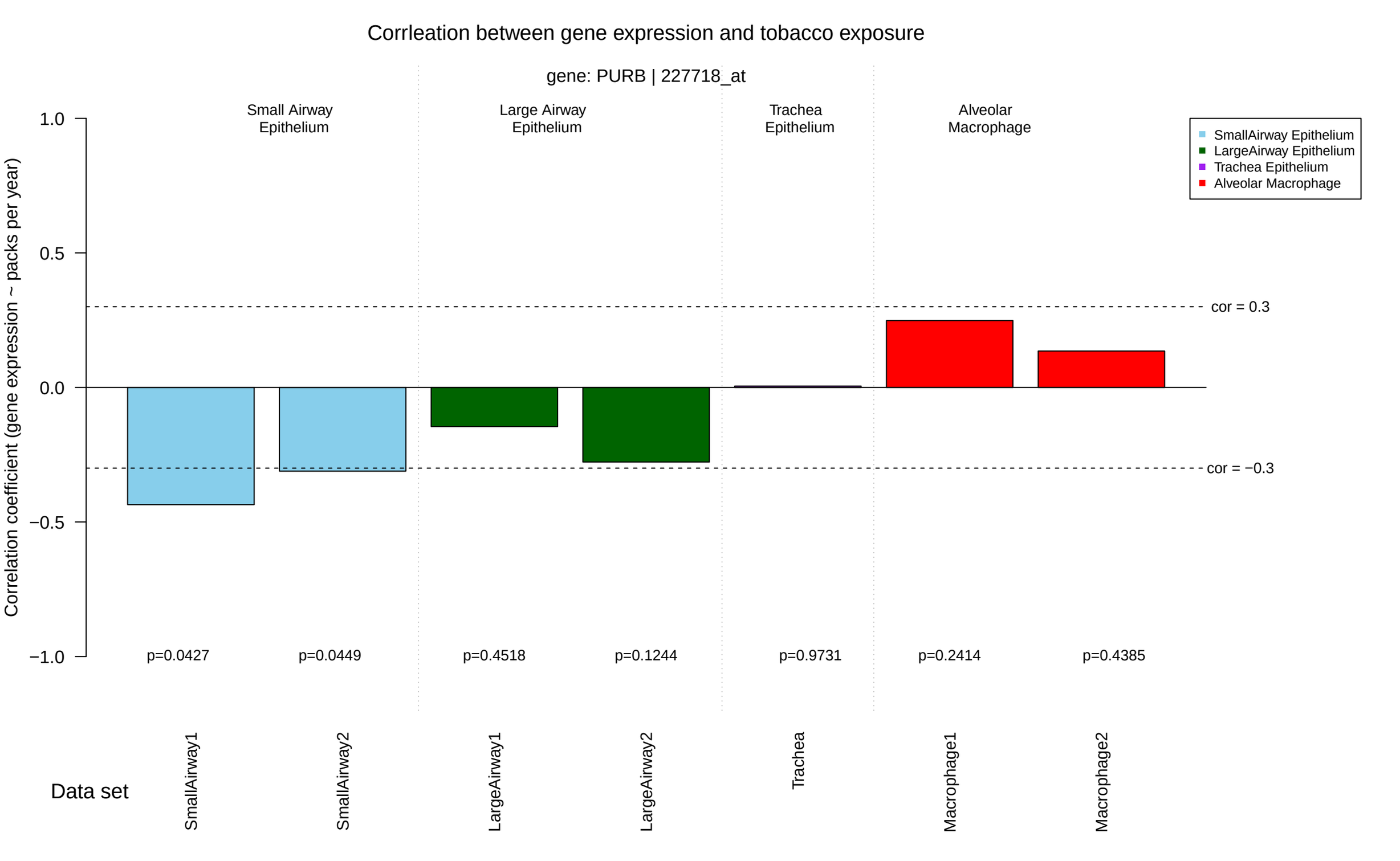

set: LargeAirway1 (p=0.1227)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.011)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5239)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1658)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5739)
PURB | 227718_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PURB | 227718_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0427
p=0.0449
p=0.4518
p=0.1244
p=0.9731
p=0.2414
p=0.4385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2